

u/10 Caching

Caching means saving a copy of data that's expensive or slow to compute / fetch, so future request can be served faster.

→ Why ?

- (-) Frontend / Browser Cache: Saves images, scripts speeds up website loads for user.
- (-) Backend / server side: In-memory cache like redis / memcached stores recent or frequent queries, computations or API results.
- (-) CDN: (Content Delivery Network): Globally caches static data close to users.
- (-) Database caching: Saves results of slow or frequent DB queries.

→ What is CDN ?

→ Content Delivery Network.

→ Deliver static content (videos, images) super fast by keeping copies near

users

(2) How does edge caching work?

→ Edge-caching means copies of frequently used data are placed on edge/CDN network servers close to the users.

Types of Caching

1) Hardware:

(i) Tiny, ultra-fast memory inside your CPU.
(L1, L2, L3, Cache)

(ii) Stores recent information/data so that CPU doesn't have to fetch from slower main RAM.

2) Software:

(i) Browser cache images

(ii) Backend system cache database

queries, API responses in memory.

(Redis / Memcached).

(iii) OS systems cache disk blocks

so that frequently access files load

faster.

3) Network Cache: It is a system that temporarily stores frequently requested sources on intermediate network.

(-) CDNs: Caches static content (videos, images) on global edge servers.

(-) Proxy Servers: Caches HTTP & related traffic to reduce bandwidth and speed up repeated requests for multiple requests.

(1) ISP: (Internet Service Provider):

Sometimes ISP cache certain popular content to serve their users.

Goal of DNS

When we type google.com, the computer needs to find numeric IP address.

Figuring out IP address for a domain name can take multiple steps and involve servers around the world. Doing

this can take a long time.

DNS lookup: Role of Caching

why?

If we look up "google.com", your computer remembers ("caches") the IP.

what happens if it's not in cache?

Computer moves to next layer: asks the DNS server (Recursive Resolver).

2) Browser Cache: Some browsers also use their own memory to save DNS lookups, so even inside your browser tabs, repeat requests are faster.

3) Recursive Resolver:

If your computer doesn't know the IP it asks the recursive resolver.

(1) The resolver sits at your ISP

(2) The resolver also caches millions of answers.

(3) Cache here:

If seen google.com it it knows,
if not it has to do the work.

Caching Strategies

- (a) Lazy caching: Don't store everything
Cache only when needed.
- (b) Write Through caching: Always keep
cache & DB in
sync.

(c) Cache Eviction Policies:

Cache is limited memory, so we need to
remove stuff that we don't need.

Most common Evictions are

- ① No limits ÷ It's limits, throws error
- ② LRU: Least Recently Used
- ③ LFU: Evict what is used least often
- ④ TTL: Remove after set time.

In Real world:

- (a) DB query caching: Reuse results for
heavy / expensive queries, so DB does less

work faster.

(b) Session String: Store temporary data (auth / session tokens) for quick look ups less DB load during authentication.

(c) Rate limiting: Track request/users in memory, act super-fast to enforce limits without DB overhead.

(d) API caching: Reuse API results, reduce external calls/costs, avoid rate limits.